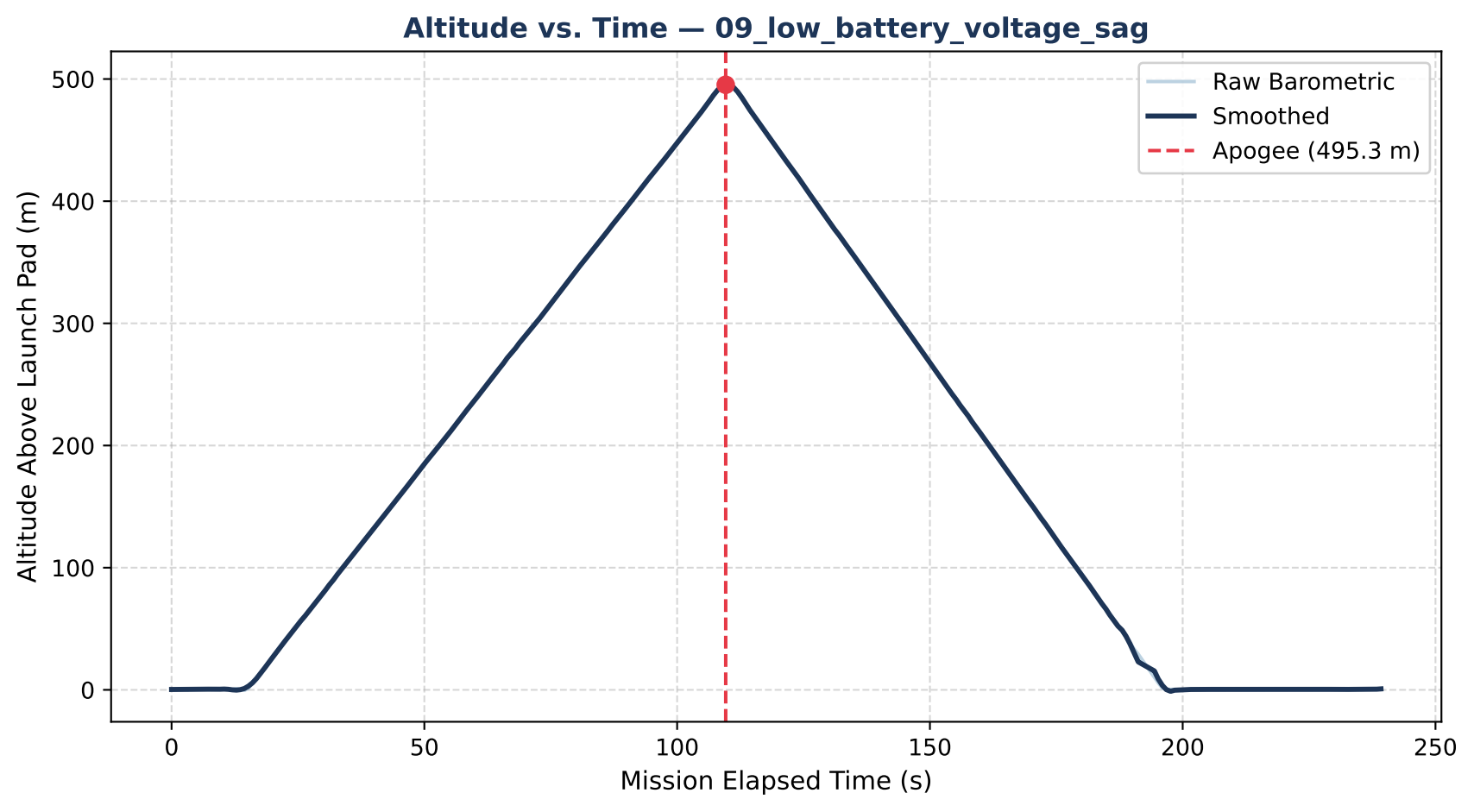
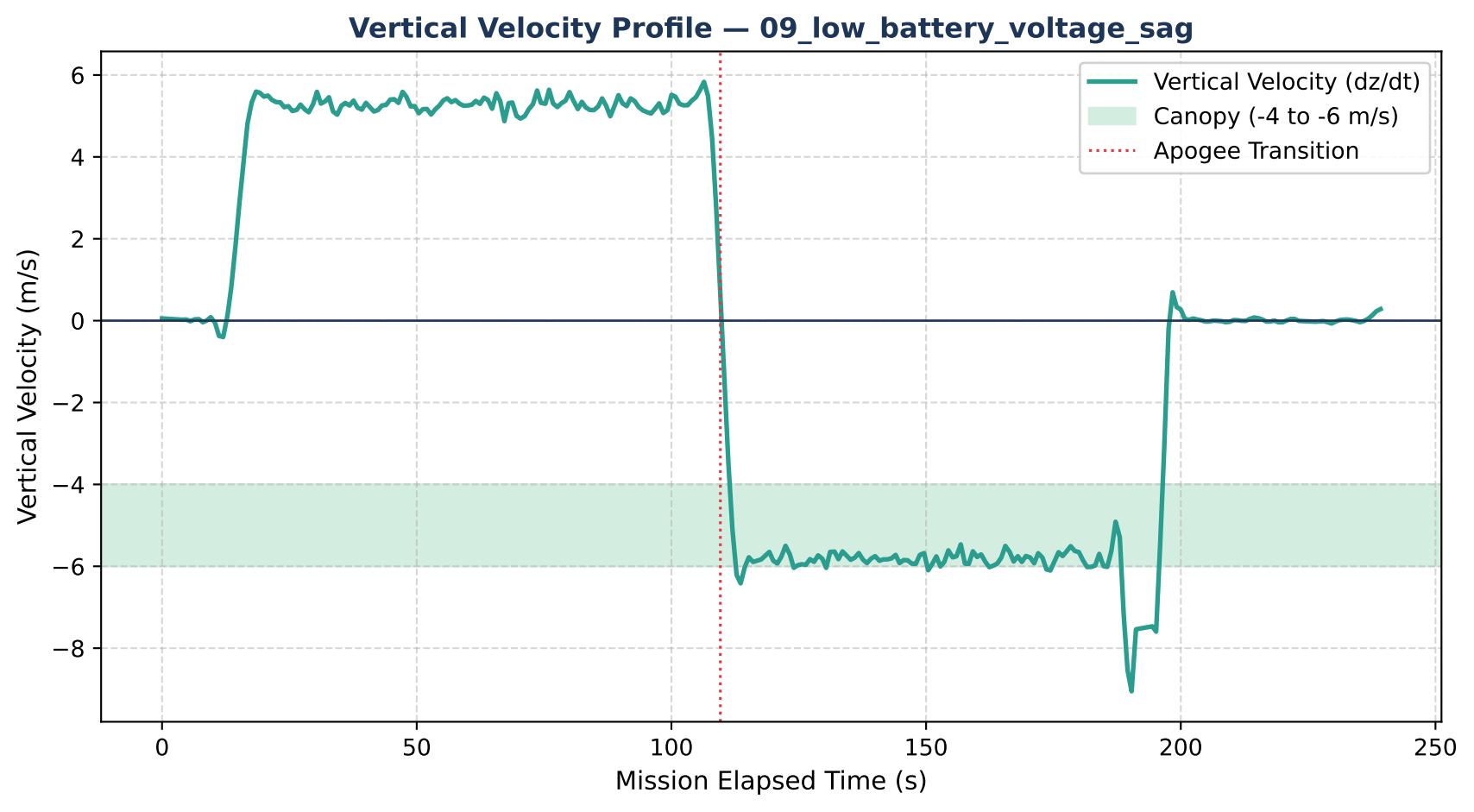
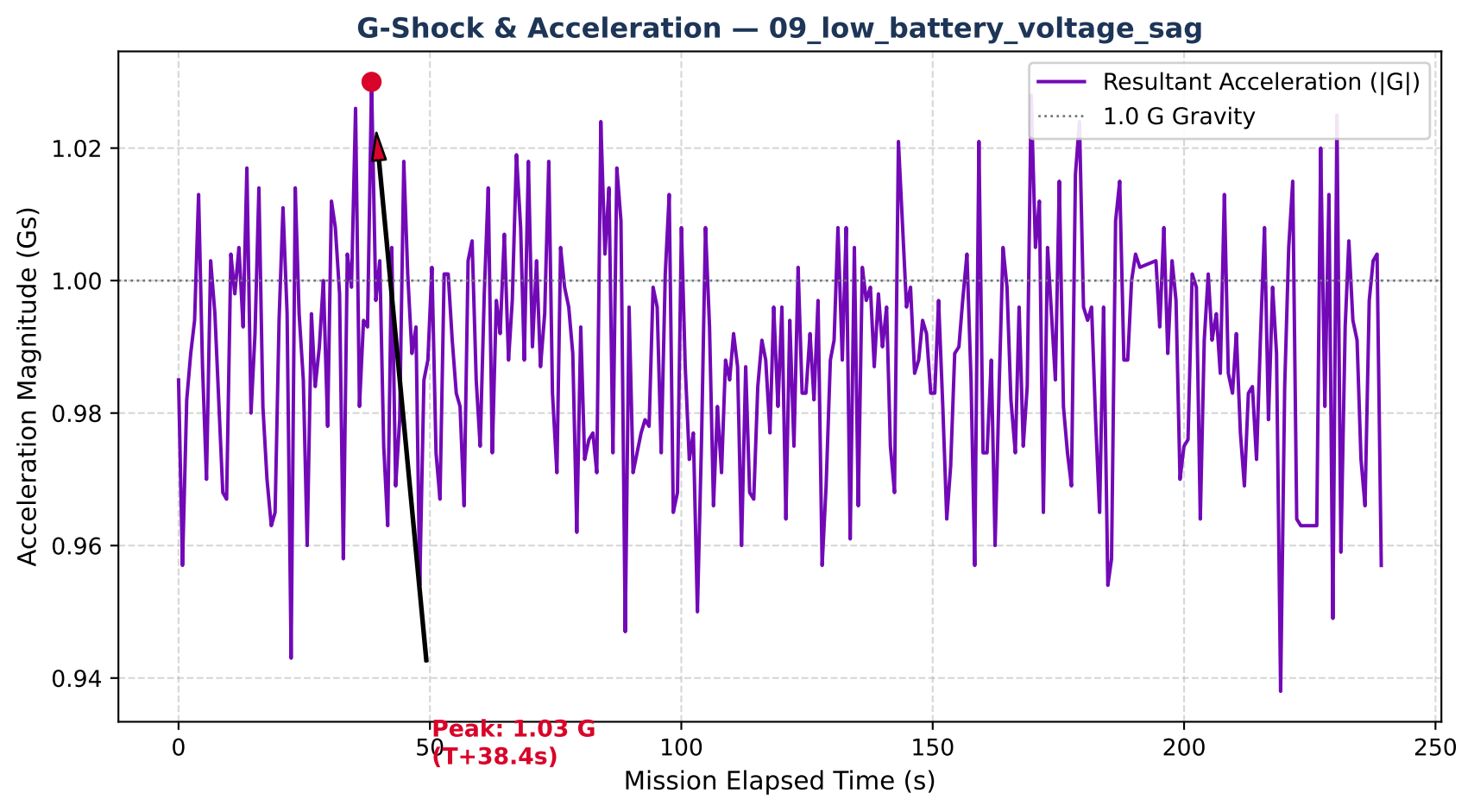
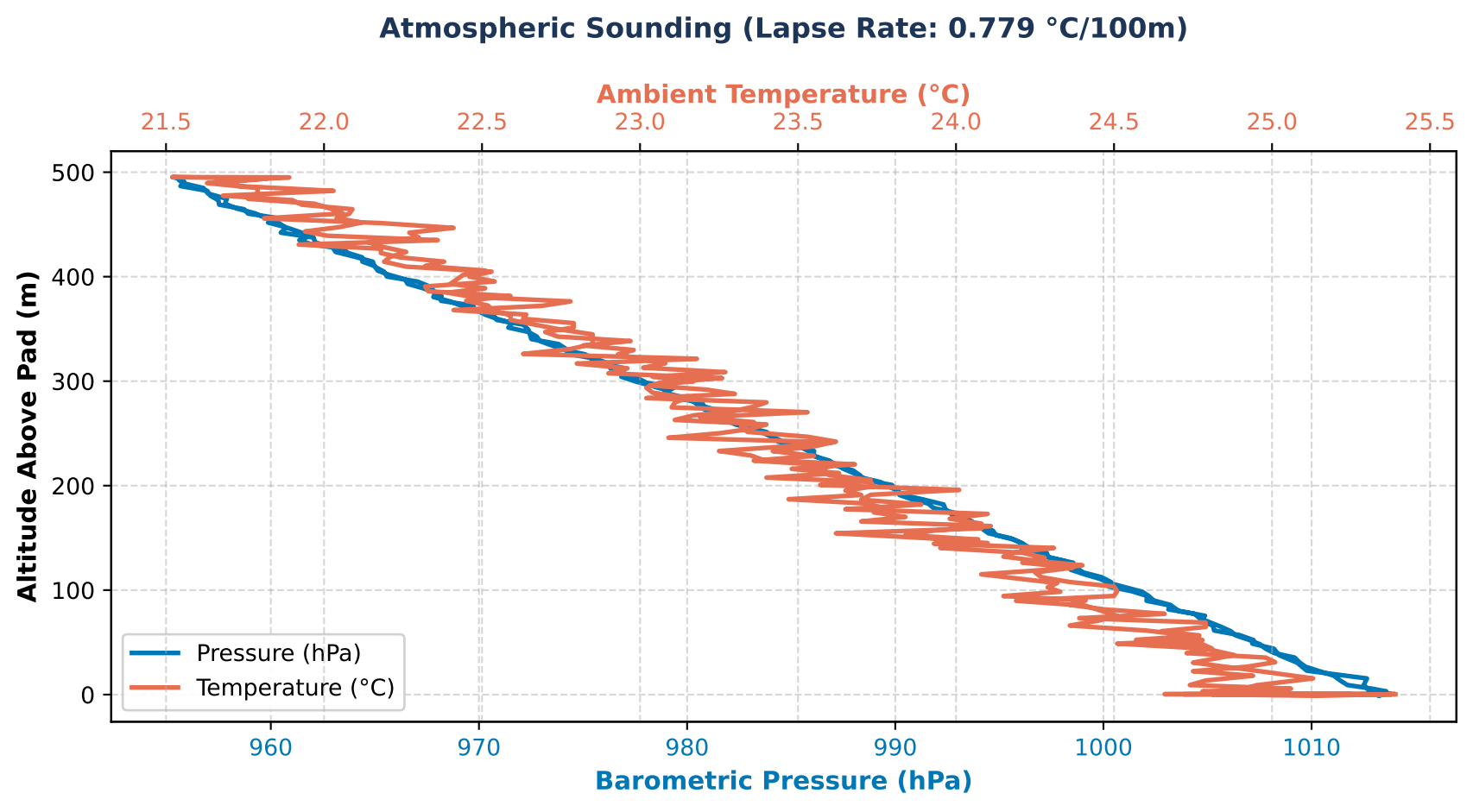
Cognitive CanSat - Mission Summary: 09_low_battery_voltage_sag

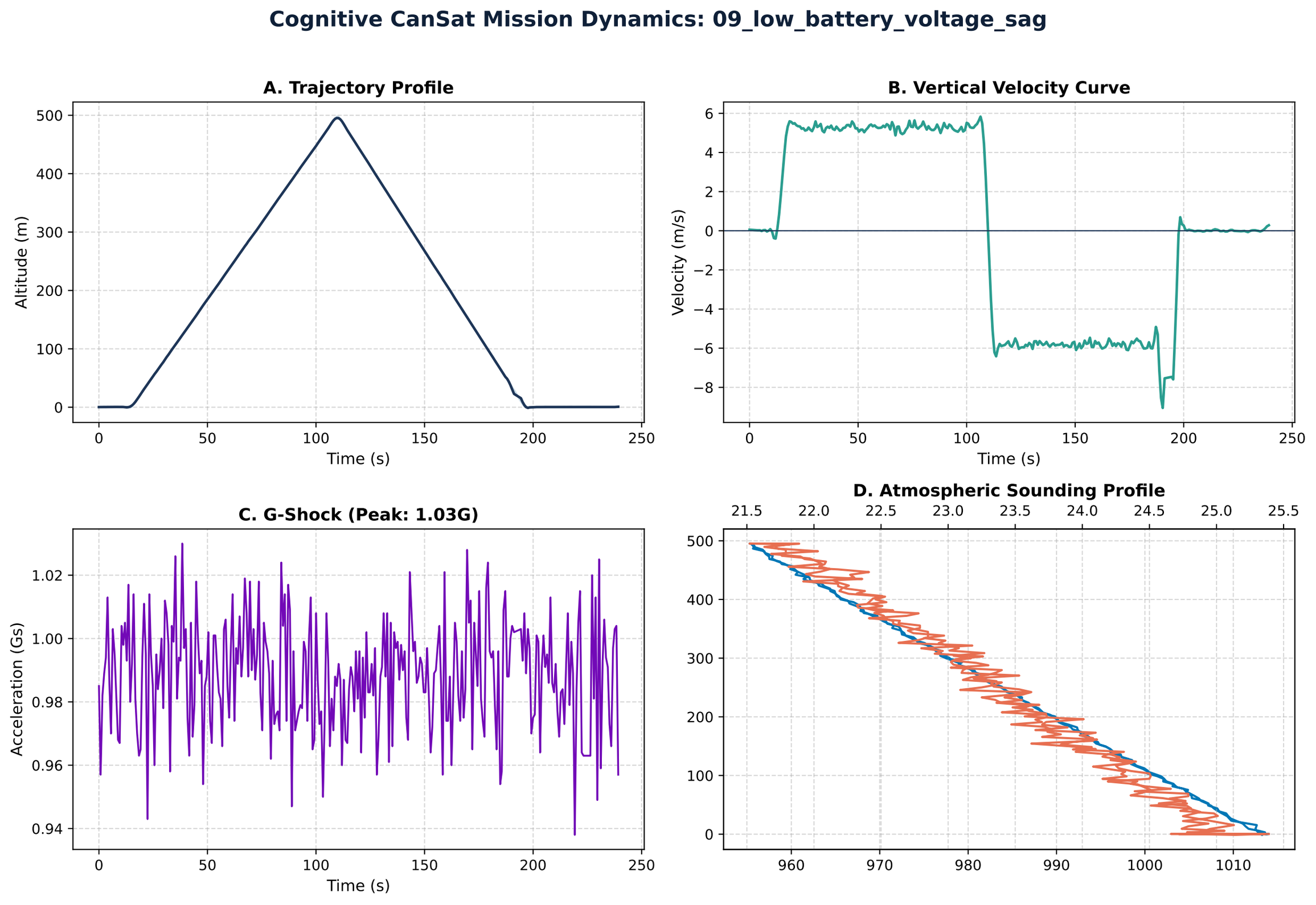
Flight Dynamics Parameter	Calibrated Value
Mission Name	09_low_battery_voltage_sag
Apogee (m)	495.29
Time to Apogee (s)	109.61
Max Ascent Velocity (m/s)	5.83
Max Descent Velocity (m/s)	-9.05
Max G-Shock (Gs)	1.03
Touchdown Shock (Gs)	1.02
Mission Duration (s)	239.22
Min Temperature (°C)	21.52
Max Temperature (°C)	25.39
Min Pressure (hPa)	955.27
Max Pressure (hPa)	1014.02
Lapse Rate (°C/100m)	0.779
Telemetry Frames	294
Landing Lat	22.574758
Landing Lon	88.367921


Altitude vs. Time — 09_low_battery_voltage_sag
Raw Barometric
Smoothed
Apogee (495.3 m)
Altitude Above Launch Pad (m)
0
100
200
300
400
500
0
50
100
150
200
250
Mission Elapsed Time (s)


Vertical Velocity Profile — 09_low_battery_voltage_sag
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
Mission Elapsed Time (s)
6
4
2
0
−2
−4
−6
−8
0
50
100
150
200
250


G-Shock & Acceleration — 09_low_battery_voltage_sag
Resultant Acceleration (|G|)
1.0 G Gravity
Acceleration Magnitude (Gs)
1.02
1.00
0.98
0.96
0.94
Peak: 1.03 G
(T+38.4s)
0
50
100
150
200
250
Mission Elapsed Time (s)


Atmospheric Sounding (Lapse Rate: 0.779 °C/100m)
Ambient Temperature (°C)
21.5
22.0
22.5
23.0
23.5
24.0
24.5
25.0
25.5
Altitude Above Pad (m)
500
400
300
200
100
0
Pressure (hPa)
Temperature (°C)
960
970
980
990
1000
1010
Barometric Pressure (hPa)


Cognitive CanSat Mission Dynamics: 09_low_battery_voltage_sag
A. Trajectory Profile
Altitude (m)
Time (s)
B. Vertical Velocity Curve
Velocity (m/s)
Time (s)
C. G-Shock (Peak: 1.03G)
Acceleration (Gs)
Time (s)
D. Atmospheric Sounding Profile